

ATTACHMENT B

MTN OBJECTIVES AND REQUIREMENTS

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Mars Telecommunications Network

Mars Telecommunications Network (MTN) Objectives and Requirements Document



**Goddard Space Flight Center
Greenbelt, Maryland**

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Table of TBDs/TBRs/TBSs

Action Item No.	Location	Summary	Individual/ Organization Actionee
TBD-01	MTN-REQ-034	Reference document for adaptive data rates currently in development.	JPL
TBD-02	MTN-REQ-049	ICD to define scientific payload	SMD

Table of Contents

1	INTRODUCTION	6
1.1	Purpose.....	6
1.2	Scope.....	6
1.3	Applicability	6
1.4	Stakeholders.....	6
2	REFERENCE AND APPLICABLE DOCUMENTS	7
2.1	Applicable Documents.....	7
2.2	Reference Documents	7
3	GOALS AND OBJECTIVES	8
4	GROUND RULES AND ASSUMPTIONS	8
5	SYSTEM OVERVIEW	9
6	MARS TELECOMMUNICATIONS NETWORK REQUIREMENTS	10
6.1	Project Requirements	10
6.2	Vehicle Requirements	15
6.3	Direct with Earth (DWE) Link Requirements	19
6.4	Proximity Link Requirements.....	23
	APPENDIX A: ABBREVIATIONS AND ACRONYMS	32
	APPENDIX B: LINK BUDGET TEMPLATE	34

List of Tables

Table 1 Ground Rules and Assumptions	9
Table 2 Science Payload Accommodation	15
Table 3 Legacy Proximity Link Spectrum.....	30
Table 4 Advanced Proximity Link spectrum	30

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1 INTRODUCTION

Consistent with law, the National Aeronautics and Space Administration (NASA) is working toward a high-performance Mars telecommunications orbiter(s), capable of providing assets in and around Mars with robust, continuous communications with earth.

1.1 Purpose

The purpose of this Objectives and Requirements Document is to provide objectives and requirements for the Mars Telecommunications Network Project.

The intent of these objectives and requirements is to support definition of the required “what” and to the maximum extent possible not define the “how”. However, since existing assets are already operating in and around Mars, legacy compatibility is required, and as such detailed specificity is provided in this document and associated applicable and reference documents. The longer-term vision is evolution of the existing Mars network to a higher performing system with extensibility to a 24/7 end state.

1.2 Scope

The scope of this requirements document includes orbiter(s), direct with earth (DWE), and Proximity link related requirements. The document does not include specific launch vehicle-related requirements. This document also describes top-level objectives of the MTN orbiter(s).

1.3 Applicability

This objectives and requirements document is applicable to the Mars Telecommunications Network Project.

1.4 Stakeholders

The following are the key stakeholders for the MTN Project:

- Exploration Systems Development Mission Directorate (ESDMD)
- Space Operations Mission Directorate (SOMD)
- Strategy and Architecture Office (SAO)
- Science Mission Directorate (SMD)
- Space Communication and Navigation (SCaN) Program
- Moon to Mars (M2M)

2 REFERENCE AND APPLICABLE DOCUMENTS

2.1 Applicable Documents

Applicable documents are those documents that form a part of this document. These documents carry the same weight as if they were stated within the body of this document.

Document Number	Document Title	Revision
<i>Architecture/Mission</i>		
NASA STD-871927	Implementing Planetary Protection Requirements for Spaceflight	
<i>Direct with Earth Links</i>		
810-007	DSN Mission Interface Design Handbook	
810-005	DSN Telecommunications Link Design Handbook	
503.0-B-2	CCSDS Tracking Data Message	
502.0-B-2	CCSDS Orbit Data Message	
<i>Proximity Links</i>		
	MRN Participation Guide	
	MRN Participation Guide Technical Support Package	
734.20-O-1	CCSDS Bundle Protocol Specification	
734.1-B-1	CCSDS Licklider Transmission Protocol (LTP)	
211.0-B-6	CCSDS Prox-1 Space Link Protocol - Data Link Layer	
732.0-B-4	CCSDS AOS Space Data Link Protocol	
732.1-B-3	CCSDS Unified Space Data Link Protocol	
211.2-B-2	CCSDS Prox-1 Space Link Protocol - Coding and Synchronization Sublayer	
211.1-B-4	CCSDS Prox-1 Space Link Protocol - Physical Layer	
NASA NPR's and Standards		
NASA-STD-7009	Standard for Models and Simulations	B
	Lunanet Interoperability Specification	5

2.2 Reference Documents

Reference documents are those documents that, though not a part of this document, serve to clarify the intent and contents of this document.

Document Number	Document Title	Revision
<i>Architecture/Mission</i>		
JPL D-106417	Mars Handbook 2022-2040	1.0
	M2M Architecture Description Document	C
NIST.FIPS.140-3	Federal Information Processing Standards Publication	
	Planetary Protection Handbook 2024	
<i>Direct with Earth Links</i>		

104-DSN	34-m BWG Stations Telecommunications Interfaces	
875-0001	DSN Mission Service Interfaces, Policies, and Practices (MSIPP)	H
820-10	Deep Space Network Services Catalog	H
<i>Proximity Links</i>		
REC 22-1	Frequency Assignment Guidelines for Communications in the Mars Region	4
NASA NPR's and Standards		
NPR 7123.001	NASA System Engineering Processes and Requirements	D

3 GOALS AND OBJECTIVES

The top-level objectives of the MTN project are identified as:

MTN-OBJ-001

Provide communication and data exchange between assets in the Mars vicinity, the Mars surface, and Earth anticipated to operate at Mars through 2035.

MTN-OBJ-002

Provide Doppler, range, and time transfer to support positioning, navigation, and timing for assets anticipated to operate at Mars through 2035.

MTN-OBJ-003

Provide communication services to existing operational missions.

MTN-OBJ-004

Provide communications, Doppler, range, and time transfer services to the Entry, Descent, and Landing (EDL) demonstration missions anticipated to operate at Mars through 2035.

MTN-OBJ-005

Provide spacecraft bus integration, accommodation, and communication services to a science payload provided by NASA.

4 GROUND RULES AND ASSUMPTIONS

The ground rules and assumptions are identified and defined in the following table:

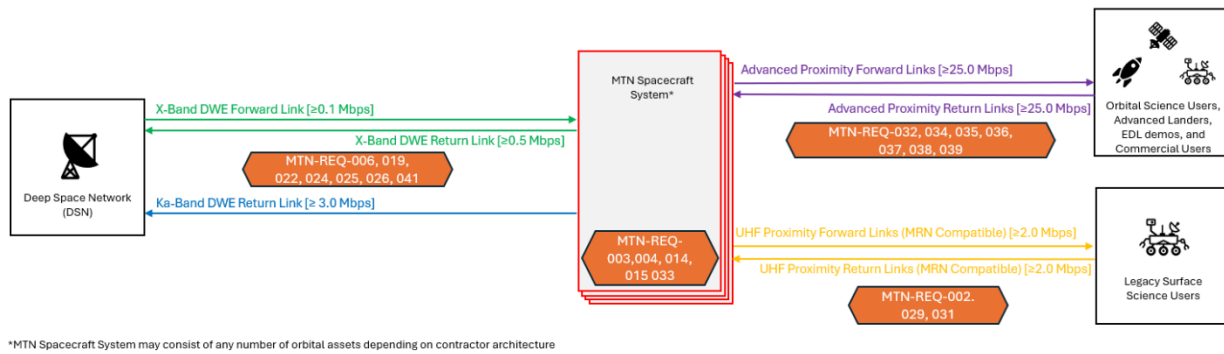
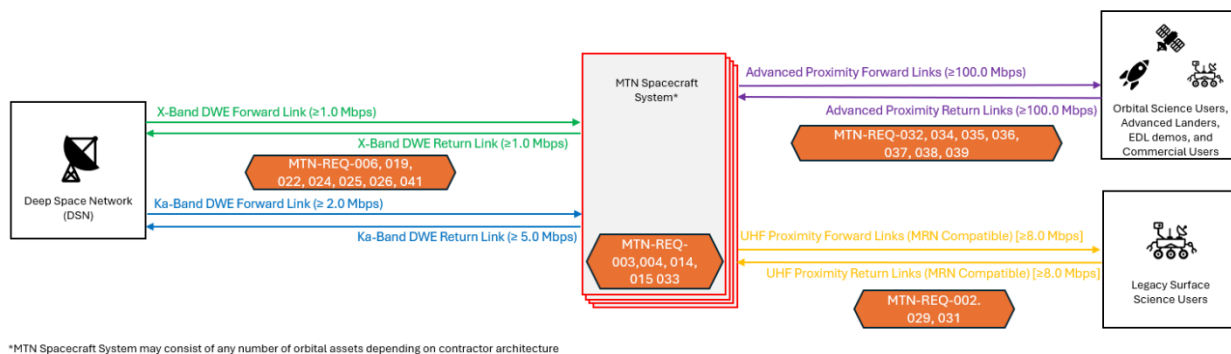
ID	Ground Rule / Assumption
1	MTN will not require, nor does it preclude surface-to-surface relay communications for 2035 uses.
2	MTN will not require, nor does it preclude orbiter crosslink capability. However, 24/7 end state extensibility is envisioned in the future.
3	The MTN will be compatible with the DSN communication links from deep space to Earth.

4	The MTN communication links from orbiter(s) to DSN will require ground compatibility testing with DSN, prior to launch. (Test assets such as DSN CTT-22, MIL-71 and DTF-21 may be available subject to negotiation with NASA)
5	The MTN contractor will deliver to NASA an MTN portable test unit for Proximity link compatibility testing for future mission users for all Proximity links.
6	MTN will require adaptive data rates on the Proximity link, consistent with existing capability.
7	MTN will not require optical communications for DWE link, but could be considered as part of a contractor's proposal
8	MTN will require low-rate X-band and higher rate Ka-band DWE connectivity throughout all operational phases.
9	MTN will require S-band or K/Ka-band Advanced Proximity links for higher rate missions
10	MTN will require UHF Legacy Proximity links.
11	NASA is procuring an MTN system via Fixed Price Contract (FFP), and scope includes ground and flight hardware delivery to NASA. Launch and future operations will be handled via contract options.
12	Launch vehicle interface will be proposed and managed by the contractor. There are no specific requirements included for a particular launch vehicle.
13	MTN will require scheduling functionality and associated interfaces for both DWE and Proximity links
14	Hosting a SMD science payload is on a do-no-harm basis, and schedule risk is critically important.

Table 1 Ground Rules and Assumptions

5 SYSTEM OVERVIEW

This document implements a nomenclature of “threshold” and “goal” requirements. The “threshold” requirements define the minimum performance and functional requirements necessary to achieve an acceptable system. The “goal” requirements define stretch targets that are desired by project stakeholders that represent meaningful improvement over the minimum viable system.

MTN Threshold Configuration**Figure 1 Mars Telecommunications Network (MTN) Threshold Configuration****MTN Goal Configuration****Figure 2 Mars Telecommunications Network (MTN) Goal Configuration****6 MARS TELECOMMUNICATIONS NETWORK REQUIREMENTS****6.1 Project Requirements****MTN-REQ-001 *Operational Lifetime***

Threshold Requirement: Mars Telecommunications Network shall provide communications and Position, Navigation, and Timing (PNT) services for a minimum operational lifetime of 5 Earth years, beginning at operational readiness.

Goal Requirement: N/A

Rationale: A 5-year operational lifetime ensures the Mars Telecommunications Network architecture can provide stable relay services across multiple Mars missions and demonstrations beginning at operational readiness. For the purposes of this requirement, operational readiness is defined as post-commissioning and ready to provide full operational capability. (Ref: Moon-to-Mars Architecture Description Document Rev C)

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-002 *Proximity Link Continuity of Service*

Threshold Requirement: Mars Telecommunications Network shall provide Proximity Link radiometric and communications services to current and in development Mars surface assets.

Goal Requirement: N/A

Rationale: Integrated communications and radiometric services are required to support commanding, telemetry, and precision navigation for existing and in-development Mars surface assets at UHF using the Proximity-1 Space Link Protocol (Ref. CCSDS 211.2-B-2, “Proximity-1 Space Link Protocol—Coding and Synchronization Sublayer”; CCSDS 211.1-B-4, “Proximity-1 Space Link Protocol - Physical Layer”; CCSDS 211.0-B-6, “Proximity-1 Space Link Protocol - Data Link Layer”) Note there are slight differences between NASA and ESA assets that must be accommodated and there are also slight differences with the protocol specification from both types of assets.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003

MTN-REQ-003 *Non-Real-Time Store-and-Forward*

Threshold Requirement: Mars Telecommunications Network shall provide store-and-forward capability for Direct with Earth (DWE) and Proximity communications.

Goal Requirement: N/A

Rationale: Providing non-real-time, store-and-forward enables reliable data delivery despite long delays, intermittent visibility, and periodic loss of line-of-sight between surface, orbital, and Earth segments ensuring continuous mission communications in Mars orbital geometries without requiring real-time connectivity.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-004 *Concurrent Service Provision*

Threshold Requirement: Mars Telecommunications Network shall support concurrent full duplex operation of communications relay DWE, Proximity communications, and associated services.

Goal Requirement: N/A

Rationale: Concurrent, non-interfering operation of communications relay, semi-real-time delivery, DTN networking, navigation, timing, and messaging is required to support overlapping mission activities without degrading critical services. This capability ensures predictable performance under peak load, protects time-critical functions, and enables Mars Telecommunications Network to operate as reliable, multi-mission infrastructure consistent with SCA architecture and NASA risk-informed systems engineering practices (CCSDS BPv7, NPR 7123.1, NPR 8705.5).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-005 *Hardware Delivery Date*

Threshold Requirement: The contractor shall deliver completed ground and flight systems to NASA including completion of all associated reviews and inspections by December 31, 2028.

Goal Requirement: N/A

Rationale: Consistent with Pub. L. 119-21, delivery by end of CY2028 ensures continuity of critical relay services and prevents a communications gap as existing Mars orbiters age.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003

MTN-REQ-006 *Direct with Earth (DWE) Service*

Threshold Requirement: Mars Telecommunications Network shall provide Direct with Earth (DWE) under all flight phases (i.e. launch, separation, transit, and insertion) excluding solar conjunction and occultations.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will provide reliable, communication and radiometric services with terrestrial ground-stations.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-007 *Mission Environment*

Threshold Requirement: Mars Telecommunications Network shall operate as specified herein when exposed to all expected environments through all expected mission phases.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network, in addition to on-orbit lifecycle, will be exposed to all relevant operational environments through deployment into orbit including but not limited to orbital insertion, time-light delays from ground command preventing timely intervention and potential intermittent loss of communication and command. [Ref. Mars Handbook 2022-2040 Release 1.0]

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-008 *Solar Conjunction*

Threshold Requirement: The Mars Telecommunications Network Direct with Earth (DWE) connections shall operate at Sun-Earth-probe angles at 3° and greater.

Goal Requirement: The Mars Telecommunications Network Direct with Earth (DWE) connections shall operate at Sun-Earth-probe angles at 2° and greater.

Rationale: The Earth-Mars solar conjunction will prevent communications between Earth and Mars for the duration the time of the actual line-of-sight blockage. Designing communication links to operate within 3° of the blockage will increase the availability of the links by incorporating mitigation techniques and link margin to optimize communications in the presence of solar adverse effects on the respective receivers. Operational constraints may be imposed to mitigate the potential of corrupted commands sent to the spacecraft. For this requirement, threshold values reflect expected performance of X-Band transmission.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-009 *Ground Commanded Prioritization*

Threshold Requirement: The Mars Telecommunications Network shall provide the capability for data prioritization.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will service multiple end users. Therefore, it is necessary to provide a means of prioritizing data to facilitate link scheduling. This prioritization scheme is controlled by ground personnel and can be specified prior to and after the network services are rendered.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-010 *Scheduling Functionality*

Threshold Requirement: The Mars Telecommunications Network shall implement a means for user scheduling and providing associated interfaces for both Direct with Earth (DWE) and Proximity links.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will service multiple end users. Therefore, it is necessary to provide for scheduling ground station assets and coordinate user scheduling of proximity interfaces along with the associated interfaces for user data.

Verification Method: Demonstration

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-011 *Link Performance*

Threshold Requirement: The Mars Telecommunications Network shall meet end-to-end performance for all communication links with ≥ 3 dB margin.

Goal Requirement: N/A

Rationale: Meeting end-to-end performance at maximum range with at least 3 dB margin ensures reliable communications under nominal conditions. The ≥ 3 dB link margin requirement provides explicit robustness against link modeling uncertainty, hardware performance drift, pointing error, aging of RF components, and unmodeled environmental or propagation effects. This aligns with NASA systems engineering guidance (e.g., NPR 7123.1 and NASA-STD-7009) to include appropriate performance margin for model-based predictions and long-life space assets, reducing the risk of link outages, increased retransmissions, or permanent loss of service. Communication links to be assessed using provided link budget template (Appendix B).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-012 *Bit Error rate*

Threshold Requirement: All Mars Telecommunications Network communication links shall have a bit error rate of 10^{-7} or better.

Goal Requirement: N/A

Rationale: This value is aligned with NASA best practice and ensures for accurate decryption of command link. Communication links to be assessed using provided link budget template (Appendix B).

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-049 *Scientific Payload Considerations*

Threshold Requirement: Mars Telecommunications Network shall accommodate a NASA science payload per [TBD-02] conforming to the interface and not-to-exceed envelope depicted in Table 2.

Criteria	Value	Units
Basic Mass	20	kg
Dimensions	55x55x45	cm x cm x cm
Nominal Power	60	W
Peak Power	60	W
Power Interface Port	28V	
Communication Interface	Wired	
Data Comm Port	RS-422, Ethernet	
Max Instantaneous Data Rate	25	kbps
Daily Data Volume	200-1000	Mb
Requires Thermal Isolation	No	
Shock Load Limit	GEVS	

Table 2 Science Payload Accommodation

Goal Requirement: N/A

Rationale: MTN stakeholders anticipate the inclusion of scientific payloads. All payloads will be negotiated with the NASA team and will not impact MTN schedule requirements. While certain attributes of Table 2 imply a payload functioning as part of the MTN flight system, a deployable CubeSat that fits within the mass and volume envelope of Table 2 is also under consideration.

Verification Method: Inspection

Traceability: MTN-OBJ-005

6.2 Vehicle Requirements

MTN-REQ-013 *Operational Readiness*

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to verify that this is the correct version prior to use.

Threshold Requirement: Mars Telecommunications Network shall achieve full operational readiness at Mars no later than end of calendar year 2030.

Goal Requirement: Mars Telecommunications Network shall achieve full operational readiness at Mars no later than end of calendar year 2029.

Rationale: Achieving Mars Telecommunications Network operational readiness by the end of 2030 ensures continuity of critical relay services and prevents a communications gap as existing Mars orbiters age. Aligning Mars Telecommunications Network's operational readiness with the 2028–2029 Mars launch opportunity reduces risk to dependent surface and orbital missions and supports stable programmatic planning and sustained Mars exploration. This is consistent with Pub. L. 119-21, to deliver the MTN orbiter(s) to NASA by end of CY20208. For the purposes of this requirement, operational readiness is defined as post-commissioning and ready to provide full operational capability.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-014 *Non-Volatile Store-and-Forward Memory (DWE)*

Threshold Requirement: Mars Telecommunications Network shall provide at least 1.0 Tbyte of non-volatile store and forward capability for the DWE link.

Goal Requirement: N/A

Rationale: Providing non-real-time data, store-and-forward capability enables reliable data delivery despite long delays, intermittent visibility, and periodic loss of line-of-sight between surface, orbital, and Earth segments ensuring reliable mission communications in Mars orbital geometries without requiring real-time connectivity. This value reflects an assumed average data-rate of 7 Mbps over an 8-hour pass per day for 2 days with a factor of 10 margin.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-015 *Non-Volatile Store-and-Forward Memory (Proximity)*

Threshold Requirement: Mars Telecommunications Network shall provide at least 0.5 Tbyte of non-volatile store and forward capability for the Proximity link(s).

Goal Requirement: N/A

Rationale: Providing non-real-time data, store-and-forward capability enables reliable data delivery despite long delays, intermittent visibility, and periodic loss of line-of-sight between

surface, orbital, and Earth segments ensuring reliable mission communications in Mars orbital geometries without requiring real-time connectivity.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-016 *Cybersecurity & Command Protection*

Threshold Requirement: Mars Telecommunications Network shall utilize cybersecurity and command link encryption practices consistent with industry best practices.

Goal Requirement: N/A

Rationale: NASA-approved security is required to protect Mars Telecommunications Network commanding, data integrity, and authentication against unauthorized access and data compromise. It is encouraged to implement existing industry practices where appropriate.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-017 *Planetary Protection Guidelines*

Threshold Requirement: Mars Telecommunications Network shall follow all Agency applicable Planetary Protection guidelines (NPD 1000.3, NASA/SP-20240016475, NASA-STD-8719.27).

Goal Requirement: N/A

Rationale: As a Class 3 Mars Orbital mission per NPR 8719.27, Mars Telecommunications Network is required to provide NASA access to the spacecraft, facilities, and documentation to accommodate planned independent verification and assurance activities, development of risk mitigation protocols, and any anomaly investigations throughout the project life cycle to achieve compliance with applicable Agency planetary protection requirements.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-018 *Orbital Debris*

Threshold Requirement: Mars Telecommunications Network shall follow all Agency applicable Mars Orbital Debris requirements (NASA-STD-8719.14, NPR 8715.6).

Goal Requirement: N/A

Rationale: Mars Telecommunications Network is required to demonstrate compliance with NASA Mars Orbital debris reduction requirements NASA-STD-8719.14 and NPR 8715.6 to limit debris generation, spacecraft vulnerability, post-mission lifetime, and maximize entry safety.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-019 *Maximum Interlink Data Transfer Rates*

Threshold Requirement: The Mars Telecommunications Network shall have the onboard capability to support continuous, minimal latency, and full rate data transfer between the Direct with Earth links and the Proximity links, and between different Proximity links.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will be capable of supporting high-rate communications on both the DWE and Proximity links. This requirement ensures that there are no hardware bottlenecks in moving data from one interface to another or in internal processing of the MTN.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-020 *Deleted*

MTN-REQ-040 *Radio Reprogramming*

Threshold Requirement: Mars Telecommunications Network shall support in-flight reprogramming of any software and firmware used either for Proximity link or DWE link protocol handling or digital signal processing.

Goal Requirement: N/A

Rationale: As a long-term relay service provider with unknown future users, it is important for MTN to support updates to the radio software and Field Programmable Gate Array (FPGA) Hardware Description Language (HDL) used for relay services, which can address both critical compatibility issues and service improvements. This has been demonstrated to be of great value with the existing Mars relay network, with post-launch upgrades significantly improving relay performance.

Verification Method: Demonstration

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-042 *Autonomous Navigation*

Threshold Requirement: N/A

Goal Requirement: MTN shall estimate its navigation state onboard without utilization of Earth-based radiometric tracking.

Rationale: MTN is chartered to support autonomous operations and onboard processing. Updating the navigation state onboard is key to enabling more autonomous operations at Mars, which will be critical to systems at Mars operating more independently from Earth.

Verification Method: Test

Traceability: N/A

6.3 Direct with Earth (DWE) Link Requirements

MTN-REQ-021 *Deleted*

MTN-REQ-022 *DWE Link 2-Way Coherent Ranging & Doppler*

Threshold Requirement: Mars Telecommunications Network shall be able to support 2-way coherent ranging and 2-way coherent Doppler on the DWE links with a range precision 1 m and Doppler precision of 0.1 mm/s (1-sigma) at a 60 second count time.

Goal Requirement: N/A

Rationale: Two-way coherent ranging and Doppler provide the highest-accuracy range and velocity measurements needed for precise navigation and orbit determination. These capabilities are essential for high-consequence operations and reduce mission risk by enabling stable, validated navigation solutions. Precision replicates current accuracy capability of existing systems.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-023 *DWE Link Communication Availability*

Threshold Requirement: Mars Telecommunications Network hardware and software shall provide an average annual DWE link availability of $\geq 95\%$, exclusive of the Earth ground segment, channel propagation effects, and occultations.

Goal Requirement: Mars Telecommunications Network hardware and software shall provide an average annual DWE link availability of $\geq 99\%$, exclusive of the Earth ground segment, channel propagation effects, and occultations.

Rationale: A service availability of at least 95% ensures Mars Telecommunications Network provides reliable, predictable support for mission-critical communications and navigation. This requirement drives resilient system design and reduces operational risk for sustained Mars exploration. This is achieved through a combination of low and high rate DWE communications. This requirement is on the Mars Telecommunications Network portion of the link and not the availability of the Earth side of the link or line of sight obstructions.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-024 *X-band DWE Link Data Rate*

Threshold Requirement: Mars Telecommunications Network shall support Deep Space X-band DWE link data rates up to at least 0.1 Mbps Forward (DSN to MTN) and up to at least 0.5 Mbps Return (MTN to DSN) at Mars–Earth maximum range (~400Mkm), using a single 34m BWG antenna. MTN shall support a maximum of 10 Mbps forward and return on the X-band DWE links at Earth-MTN ranges where sufficient link margin exists to meet MTN-REQ-012.

Goal Requirement: Mars Telecommunications Network shall support Deep Space X-band DWE Link data rates up to at least 1.0 Mbps Forward (DSN to MTN) and up to at least 1.0 Mbps Return (MTN to DSN) at Mars–Earth maximum range (~400Mkm), using a single 34m BWG antenna.

Rationale: Specifying minimum DWE data rates at Mars–Earth maximum range ensures Mars Telecommunications Network can provide reliable commanding and high-volume data return under worst-case link conditions. Specified rates support efficient, high-throughput operations consistent with DSN services. Designing to these thresholds drives DSN compatibility and supports resilient, campaign-level Mars Telecommunications Network infrastructure in accordance with DSN interface guidance, SCaN architectural direction, and NASA systems engineering practices (DSN MIDH 810-007; DSN IRD; NPR 7123.1; Mars Telecommunications Network Handbook). Communication links to be assessed using provided link budget template (Appendix B).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-041 *Ka-band DWE Link Data Rate*

Threshold Requirement: Mars Telecommunications Network shall support Deep Space Ka-band DWE Link data rates up to at least 3.0 Mbps Return (MTN to DSN) at Mars–Earth maximum

range (~400Mkm), using a single 34m BWG antenna. MTN shall support a maximum of 50 Mbps return on the Ka-band DWE links at Earth-MTN range where sufficient link margin exists to meet MTN-REQ-012.

Goal Requirement: Mars Telecommunications Network shall support Deep Space Ka-band DWE Link data rates up to at least 2.0 Mbps Forward (DSN to MTN) and up to at least 5.0 Mbps Return (MTN to DSN) at Mars–Earth maximum range (~400Mkm), using a single 34m BWG antenna.

Rationale: Specifying minimum DWE data rates at Mars–Earth maximum range ensures Mars Telecommunications Network can provide reliable commanding and high-volume data return under worst-case link conditions. Specified rates support efficient, high-throughput operations consistent with DSN services. Designing to these thresholds drives DSN compatibility and supports resilient, campaign-level Mars Telecommunications Network infrastructure in accordance with DSN interface guidance, SCaN architectural direction, and NASA systems engineering practices (DSN MIDH 810-007; DSN IRD; NPR 7123.1; Mars Telecommunications Network Handbook). Communication links to be assessed using provided link budget template (Appendix B).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-025 *DWE Link / Deep Space Network (DSN) Compatibility*

Threshold Requirement: Mars Telecommunications Network shall be compatible with DSN per DSN Mission Interface Design Handbook (810-007) and DSN Telecommunications Link Design Handbook (810-005).

Goal Requirement: Mars Telecommunications Network shall be compatible with DSN per DSN Mission Interface Design Handbook (810-007), DSN Telecommunications Link Design Handbook (810-005) and provide functionality to support Multiple Users per Aperture (MUPA)

Rationale: Compatibility with the Deep Space Network (DSN) ensures that Mars Telecommunications Network can use NASA's primary interplanetary communications infrastructure without custom interfaces or redesign. The DSN Mission Interface Design Handbook (810-007) is the authoritative source for all flight projects using the DSN and provides required interface guidance, service capabilities, and customer requirements. Designing to 810-007 reduces technical and schedule risk, ensures interoperable communications and tracking services, and enables clear verification against an approved DSN baseline. Multiple Users per Aperture (MUPA) is a desired functionality which will require the ability to acquire an uplink signal when multiple spacecraft are using a single aperture.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-026 ***DWE Link / CCSDS Protocol Compatibility***

Threshold Requirement: Threshold Requirement: Mars Telecommunications Network shall implement the CCSDS Bundle Protocol version 7 (BPv7) (ref. CCSDS 734.20-O-1) for data transfer in the DWE link.

Goal Requirement: Mars Telecommunications Network shall implement the CCSDS Licklider Transmission Protocol (LTP) (ref. CCSDS 734.1-B-1), for reliable data transfer in the DWE link.

Rationale: Implementing CCSDS BPv7-compliant interfaces ensures Mars Telecommunications Network can reliably interoperate with NASA, international, and commercial assets using standardized communications protocols.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-044 ***DWE Return Link Selectable Data Rates***

Threshold Requirement: The Mars Telecommunications Network DWE Return Link shall support a series of selectable data rates to efficiently utilize the available link margin across the variable Mars-Earth range.

Goal Requirement: N/A.

Rationale: As the distance between Mars and Earth varies from approximately 55 million to 400 million kilometers over their respective orbits, the associated change in free-space path loss produces substantial variation in signal strength at Earth. When the planets are nearer, the reduced path loss enables higher data rates and yet still maintains required link margins. Adapting data rates throughout the orbital cycle ensures efficient and robust communication across all DWE frequencies.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-045 ***DWE Forward Link Selectable Data Rates***

Threshold Requirement: The Mars Telecommunications Network DWE Forward Link shall support a series of selectable data rates to efficiently utilize the available link margin across the variable Mars-Earth range.

Goal Requirement: N/A.

Rationale: As the distance between Mars and Earth varies from approximately 55 million to 400 million kilometers over their respective orbits, the associated change in free-space path loss produces substantial variation in signal strength at Earth. When the planets are nearer, the reduced path loss enables higher data rates and yet still maintains required link margins. Adapting data rates throughout the orbital cycle ensures efficient and robust communication across all DWE frequencies.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-047 *DWE Link Channel Assignment*

Threshold Requirement: The Mars Telecommunications Network DWE links shall operate on the frequency channels/bandwidths assigned by NASA.

Goal Requirement: N/A.

Rationale: The MTN does not expect bandwidth for x-band to exceed 10MHz and ka-band to exceed 100MHz. Frequency authorization for each communication link is needed in accordance with NPD 2570.5E NASA RF Spectrum Management and National Telecommunications and Information Administration (NTIA) Manual of Regulations & Procedures for Federal Radio Frequency Management (May 2013 Edition, May 2014 Revisions) Chapter 10.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

6.4 Proximity Link Requirements

MTN-REQ-027 *Proximity Link Orbital Coverage*

Threshold Requirement: The Mars Telecommunications Network shall provide service to surface assets and orbital assets up to an altitude of 500 km.

Goal Requirement: N/A

Rationale: This requirement serves to set the service minimum service volume for orbital users of The Mars Telecommunications Network.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-028 *Proximity Link Surface Coverage*

Threshold Requirement: Mars Telecommunications Network shall provide services for all latitudes up to $\pm 50^\circ$ from the Martian geographical equator, and for all longitudes.

Goal Requirement: N/A

Rationale: Near-global coverage ensures the Mars Telecommunications Network system can provide reliable services for all latitudes up to ± 50 degrees from the Martian geographical equator, and for all longitudes, with only limited polar exclusions. This capability supports current NASA missions and reduces operational risk by providing consistent, predictable relay access aligned with sustained, campaign-level Mars exploration objectives. If implementing a multi-spacecraft system, this requirement applies to the network, not each unit. Coverage should optimize number of passes and pass duration to extent possible within the design constraints. For reference, current surface assets receive 2 passes per sol with current relay systems.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-029 *Proximity Link Time Transfer Data Correlation*

Threshold Requirement: Mars Telecommunications Network shall provide radiometric time-transfer services to surface and orbital assets where the time correlation accuracy of in-situ time transfer data is traceable to Coordinated Universal Time (UTC) to within ≤ 1 millisecond error (3-sigma) at the user interface.

Goal Requirement: The Mars Telecommunications Network shall provide radiometric time-transfer services to surface and orbital assets where the time correlation accuracy of in-situ time transfer data is traceable to Coordinated Universal Time (UTC) to within ≤ 1 microsecond error (3-sigma) at the user interface.

Rationale: Accurate radiometric time-transfer provides timing data needed for precise positioning, navigation, and timing at Mars. This capability reduces navigation errors, supports time-critical operations, and enables reliable, interoperable PNT services for surface and orbital assets. This requirement is not intended for specifying formation of a Mars time scale (at least at this time) that is maintained to UTC or imply that Mars timing should utilize leap seconds; however, it is intended to specify the capability of the MTN to correlate its own clocks and time relative to UTC (the civilian timescale we use and 'tied' to Earth Terrestrial Time (TT)) and then be able to transfer that time correlation information to user clocks so that they can relate their time to Earth-based times. Note that this will require monitoring and tracking the physical drift of MTN's clocks and associated frequencies as well as the proper time/frequency expected in the Mars local relativistic environment (and how it relates to terrestrial and solar system barycentric relativistic environments).

Verification Method: Test

Traceability: MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-030 *Proximity Link Communication Availability*

Threshold Requirement: Mars Telecommunications Network shall provide an average annual Proximity Link service availability $\geq 98\%$ during the Proximity Communications windows.

Goal Requirement: Mars Telecommunications Network shall provide an average annual Proximity Link service availability $\geq 99\%$ during the Proximity Communications windows.

Rationale: A service availability of at least 98% ensures Mars Telecommunications Network provides reliable, predictable support for mission-critical communications and navigation. This requirement drives resilient system design and reduces operational risk for sustained Mars exploration.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-031 *UHF Proximity Link Data Rates*

Threshold Requirement: Mars Telecommunications Network shall support UHF Proximity Link communications services for surface and orbital assets at Mars with per-user data rates up to at least 2 Mbps under nominal conditions at zenith, with the zenith distance defined to be the closest approach radial distance of MTN to the surface or orbital asset.

Goal Requirement: Mars Telecommunications Network shall support UHF Proximity Link communications services for surface and orbital assets at Mars with per-user data rates up to at least 8 Mbps under nominal proximity rate conditions at zenith, with the zenith distance defined to be the closest approach radial distance of MTN to the surface or orbital asset.

Rationale: Mars surface and orbital missions depend on Proximity relay communications to return science, engineering, and health data when direct-to-Earth links are unavailable. Existing NASA Mars missions and relay orbiters establish a proven operational baseline for UHF Proximity return-link throughput that the Mars Telecommunications Network seeks to build upon. Specifying minimum Proximity return-link data rates ensures continuity and interoperability with deployed Mars assets while preserving flexibility in Proximity relay architecture and implementation. Typically, the lower bound on the communications distance (the zenith distance in the requirement) will have a lower bound equal to the minimal orbital altitude of the MRN. Communication links to be assessed using provided link budget template (Appendix B).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003

MTN-REQ-032 *Advanced Proximity Link Data Rates*

Threshold Requirement: Mars Telecommunications Network shall support Advanced (non-UHF) Proximity Link communications for surface and orbital assets at Mars with per-user data rates up to at least 25 Mbps under nominal Proximity rate conditions at zenith, with the zenith distance defined to be the closest approach radial distance of MTN to the surface or orbital asset.

Goal Requirement: Mars Telecommunications Network shall support Advanced (non-UHF) Proximity Link communications for surface and orbital assets at Mars with per-user data rates up to at least 100 Mbps under nominal Proximity rate conditions at zenith, with the zenith distance defined to be the closest approach radial distance of MTN to the surface or orbital asset.

Rationale: Mars surface and orbital missions depend on Proximity relay communications to return science, engineering, and health data when direct-to-Earth links are unavailable. Advanced Proximity link data rates are derived from stakeholder input and anticipated needs within the MNT operational lifetime. Typically, the lower bound on the communications distance (the zenith distance in the requirement) will have a lower bound equal to the minimal orbital altitude of the MRN. Communication links to be assessed using provided link budget template (Appendix B).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-033 *Simultaneous Proximity Links*

Threshold Requirement: Mars Telecommunications Network shall simultaneously support ≥ 2 simultaneous concurrent full-duplex legacy links and ≥ 1 advanced Proximity links within the Mars Telecommunications Network Proximity antenna fields of view.

Goal Requirement: N/A

Rationale: Supporting at least two concurrent full-duplex legacy links and at least one advanced proximity link (for a total of at least 3 links) enables Mars Telecommunications Network to provide simultaneous two-way communications for multiple surface and orbital users, reducing scheduling conflicts and supporting time-critical operations. This capability ensures Mars Telecommunications Network can function as shared, resilient campaign infrastructure with acceptable operational risk (NASA-HDBK-1002).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-034 *Adaptive Data Rates*

Threshold Requirement: The Mars Telecommunications Network Proximity link system shall support adaptive data rates from surface and orbital assets.

Goal Requirement: N/A

Rationale: Adaptive Proximity links will achieve significant improvement in throughput capacity by using available link margin. Requires basic changes to the transmitter and/or receiver parameters including modulation, symbol rates, frequency channels, symbol coding, and others that affect the throughput and performance of the communications channels when conditions change. (ref. CCSDS 211.0-B-6, TBD-01)

Verification Method: Analysis

Traceability: MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-035 *Proximity Link 1-Way Doppler*

Threshold Requirement: The Mars Telecommunications Network shall provide in situ 1-way Doppler tracking services to users with a precision of 0.5mm/s (1-sigma) at a 60 second count time.

Goal Requirement: The Mars Telecommunications Network shall provide in situ 1-way Doppler & Ranging services to users with a precision of 0.5mm/s (1-sigma) at a 60 second count time and ranging precision of 1 meter.

Rationale: Pseudo-range and one-way forward Doppler reference signals support navigation when two-way tracking is unavailable or impractical. These services support high-dynamics mission phases, reduce onboard complexity for users, and improve navigation continuity, autonomy, and overall mission resilience.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-036 *Proximity Link 2-Way Doppler*

Threshold Requirement: The Mars Telecommunications Network shall provide for in situ 2-Way coherent Doppler tracking services to users with a precision of 0.5mm/s (1-sigma) at a 60 second count time with a latency of less than 10 seconds.

Goal Requirement: The Mars Telecommunications Network shall provide for in situ 2-way coherent Doppler & Ranging services to users with a precision of 0.5mm/s (1-sigma) at a 60 second count time and ranging precision of 1 meter with a latency of less than 10 seconds.

Rationale: Two-way coherent ranging and Doppler provide higher accuracy measurements (relative to one way measurements) needed for precise navigation and orbit determination. These

capabilities are essential for high-consequence operations and reduce mission risk by enabling stable, validated navigation solutions. Precision replicates current accuracy capability of existing systems.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-037 *Position, Navigation, and Timing (Service Reference Signals)*

Threshold Requirement: The Mars Telecommunications Network shall provide pseudorange (PR) and 1-way forward Doppler reference signals with a signal-in-space-errors (SISE) contributions of root sum square (RSS) position and velocity errors, respectively, of ≤ 100 meters and 5 cm/sec (3-sigma).

Goal Requirement: The Mars Telecommunications Network shall provide pseudorange (PR) and 1-way forward Doppler reference signals with a signal-in-space-errors (SISE) contributions of root sum square (RSS) position and velocity errors, respectively, of ≤ 50 meters and 3 cm/sec (3-sigma).

Rationale: The Mars Telecommunications Network shall be capable of providing reference signals that can be used by all assets at any time to calculate their own position and time. SISE includes errors from: spacecraft on-board state knowledge (position, velocity and time), signal integrity and on-board path delays, predicted spacecraft states conveyed to users, and states and time representation. SISE does not include delays and phase noise introduced by the user system. The threshold value reflects what is achievable utilizing procurable USO clocks. The Goal value assumes use of atomic clock system.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-038 *Advanced Proximity Link Protocol Standard*

Threshold Requirement: Mars Telecommunications network Advanced S-band or K/Ka-band proximity links shall comply with Section 4.2 LNSP-User Proximity Interfaces and Section 4.4 LNSP-User Contingency Interfaces in the Lunanet interoperability Specification, Rev 5, and all Advanced Proximity Links band shall implement a comparable interoperable (i.e. not proprietary) interface compliant with appropriate CCSDS protocol standards.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will implement a non-proprietary standard for Proximity link protocol.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-039 *Open-Loop Recording*

Threshold Requirement: The Mars Telecommunications Network shall provide open-loop recording for critical events (of 10 minutes in duration or less) at a selectable sampling rate ≤ 2 MHz with ≥ 8 bits per each component of a complex sample including recording at least 8 bits of AGC information and a 32-bit time tag at a rate ≥ 1 Hz.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network shall support the needed functionality currently available to legacy users during critical events (e.g. EDL, Mars Ascent, etc) for all operational proximity bands.

Verification Method: Test

Traceability: MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-043 *Service Provision to Surface Assets*

Threshold Requirement: The Mars Telecommunications Network shall support communications and navigation with any surface asset in view of the Mars Telecommunications Network, when above 10 degrees elevation angle, measured from the surface asset to Mars Telecommunications Network.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network shall provide service to all surface users within view of the system. For the purposes link assessment, field of view below 10 degrees of elevation will not be supported.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003

MTN-REQ-046 *Legacy Proximity Link Spectrum*

Threshold Requirement: The Mars Telecommunications Network Legacy Proximity Links shall operate at UHF frequency from Table 3.

Label	Link Direction	Frequency
Proximity Forward	Orbit-to-User:	435-450 MHz
Proximity Return	User-to-Orbit	390-405 MHz

Table 3 Legacy Proximity Link Spectrum

Goal Requirement: N/A

Rationale: The proximity link frequency selection supports interoperability from different service providers for legacy and future user spacecraft, meet internationally coordinated spectrum, and where applicable, enables leveraging hardware and standards' developments for the moon that may be used for Mars (e.g. LNIS, CLPS/CMPS, human class landers, rovers, habitats, payloads). These values are aligned with the SFCG Recommendation, SFCG 22-1R4 and International Table of Frequency Allocations for Mars. Note that these frequencies are currently SFCG recommendations only and do not guarantee allocation for Mars Proximity applications.

Verification Method: Test

Traceability: MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-048 *Advanced Proximity Link Spectrum*

Threshold Requirement: The Mars Telecommunications Network Advanced Proximity links shall operate at a minimum of either S-band or K/Ka-band frequency from Table 4 for User to Orbit and Orbit to User links.

Label	Link Direction	Frequency
Proximity Forward	Orbit-to-User:	2025-2110 MHz 22.55-23.55 GHz
Proximity Return	User-to-orbit	2200-2300 MHz 25.5-27.5 GHz

Table 4 Advanced Proximity Link spectrum

Goal Requirement: The Mars Telecommunications Network Advanced proximity links shall operate at a minimum of K/Ka-band frequency from Table 4 for User to Orbit and Orbit to User links.

Rationale: The proximity link frequency selection supports interoperability from different service providers for legacy and future user spacecraft, meet internationally coordinated spectrum, and where applicable, enables leveraging hardware and standards' developments for the moon that may be used for Mars (e.g. LNIS, CLPS/CMPS, human class landers, rovers, habitats, payloads). These values are aligned with the SFCG Recommendation, SFCG 22-1R4 and International Table of Frequency Allocations for Mars. Note that these frequencies are currently SFCG recommendations only and do not guarantee allocation for Mars Proximity applications. Bandwidth use is expected to be spectrum-efficient within the specified bands and coordinated with NASA.

Verification Method: Test

Traceability: MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-050 *Raw Data Mode*

Threshold Requirement: The Mars Telecommunications Network shall provide the ability to receive, store, and forward demodulated and decoded bits directly as received without any protocol applied for all proximity links.

Goal Requirement: N/A

Rationale: This mode is called “raw data mode” and is helpful during EDL events to capture and quickly return data transmitted from these incoming Mars missions without requiring post-processing by ground operators, which is typical of open-loop recordings (see MTN-REQ-039). This mode is not specified in the Proximity-1 specification and is distinct from Proximity-1’s “expedited mode”, which requires that received data be structured as Prox-1 frames and rejects frames with bad CRCs. The use of this mode does not preclude the use of the open-loop recording of the same received signal.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003

Appendix A: Abbreviations and Acronyms

AGC	Automatic Gain Control
BER	Bit Error Rate
BP	Bundle Protocol
BWG	Beam Wave Guide
CCSDS	Consultative Committee for Space Data Systems
CLTU	Communications Link Transmission Unit
CY	Calendar Year
dB	Decibel
DSN	Deep Space Network
DWE	Direct with Earth
EDL	Entry, Descent, and Landing
EIRP	Effective Isotropic Radiated Power
ESDMD	Exploration Systems Development Mission
FPGA	Field Programable Gate Array
FWD	Forward
GEVS	General Environmental Verification Standard
HDL	Hardware Description Language
IRD	Interface Requirements Document
kHz	Kilohertz
km	Kilometer
LDPC	Low-Density Parity-Check
m	meter
M2M	Moon to Mars
Mbps	Megabits per Second
mm	Millimeter
MTN	Mars Telecommunications Network
MUPA	Multiple Users Per Aperture
NASA	National Aeronautics and Space Administration
NPR	NASA Procedural Requirements
PNT	Position, Navigation, and Timing
RTN	Return
s	Second
SAO	Strategy and Architecture Office
SCaN	Space Communication and Navigation
SISE	Signal-in-Space-Error
SMD	Science Mission Directorate
SOMD	Space Operations Mission Directorate

Tbyte	Terabyte
UHF	Ultra-High Frequency
USO	Ultra Stable Oscillator
UTC	Coordinated Universal Time

Released Version

Appendix B: Link Budget Template

The following link budget templates are provided for the vendor to fill out as part of their proposal. List all gains and margins as positive and all losses as negative in the link budgets provided. These templates provide initial insight into a representative approach and are not constraints on the links the vendor can propose or implement. Include link budgets for all links at all frequencies for a proposed configuration.

Direct With Earth (DWE) Links

DWE Return Link at Max Distance (MTN to DSS-55)						
	Node	Parameter	Value	Units	Equation	Comments
1	MTN	Transmit Power		dBW	Input, Antenna Dependent	e.g. Circuit Losses, Antenna Pointing Losses ¹
2	MTN	Transmit Losses		dB	Input, Antenna Dependent	
3	MTN	S/C Antenna Efficiency		%	Input, Antenna Dependent	
4	MTN	S/C Antenna Diameter		m	Input, Antenna Dependent	
5	MTN	S/C Antenna Gain		dB	$10*\log_{10}[(3)*(4)*(8)*\pi*1000000/c]^2]$	
6	MTN	EIRP		dBW	(1)+(2)+(5)	
7	Channel	Distance		km	Input, Orbit Dependent ²	
8	Channel	Center Frequency		MHz	Input, Antenna Dependent	
9	Channel	Free Space Loss		dB	$-[20*\log_{10}[(7)]+20*\log_{10}[(8)]+32.45]$	ITU-R P.525-5
10	Channel	Total Atmospheric Loss		dB	Input -0.5 dB at X-band or -5 dB at Ka-band	Atmospheric Loss at DSS-55, MDSCC
11	Channel	Other Losses		dB	Input, Antenna Dependent	Polarization/RFI Losses ¹
12	DSN	Power Arriving at Antenna		dB	(6)+(9)+(10)+(11)	
13	DSN	Receive G/T		dB/K	Input 54.2 dB/K at X-band or 61.1 dB at Ka-band	DSN 820-100-H
14	DSN	Received P_r/N_0		dB-Hz	(12)+(13)- $10*\log_{10}[1.38E-23]$	K (Boltzmann's Constant) = 1.38e-23 W/K/Hz
15		Carrier Suppression		dB	Input, Losses for Carrier Synchronization	
16		Ranging Suppression		dB	Input, Losses for Simultaneous Ranging	
17	DSN	Received P_d/N_0		dB-Hz	(14)-(15)-(16)	
18	DSN	Implementation Loss	-1	dB	See Note ³	DSN 810-005
19		Receive Symbol Rate (Linear)		Msp/s	Input, Modulation Dependent	
20		Receive Symbol Rate (dB)		dB-Hz	$10*\log_{10}[(19)*1000000]$	
21		Coding Rate		-	Input Coding Rate ⁴	
22		Receive Data Rate		Mbps	(19)*(21)	
23		Required E_b/N_0		dB	Input, Dependent on Coding Scheme ⁵	MTN-REQ-012: BER = 10^{-7}
24		Required E_r/N_0		dB	(23)+ $10*\log_{10}[(21)]$	
25		Required Margin	3	dB		MTN-REQ-011: Margin ≥ 3 dB
26		Excess Margin		dB	(17)+(18)-(20)-(24)-(25)	

Table B-1: DWE Return Link

¹List out all losses that are covered within these rows.

²Provide link budgets for both the minimum and maximum distances between Mars and Earth, as listed in **MTN-REQ-044**. Use 55,000,000 km for the minimum Mars-Earth Distance and 400,000,000 km for the maximum Mars-Earth Distance.

³The modulations compatible with DSN are documented in the DSN Link Design Handbook, 810-005 Module 206. If using GMSK at rates higher than 10 Mbps, assume an extra 1 dB implementation loss at the DSN receiver.

⁴Select a DSN-compatible coding scheme and list the rate in this row.

⁵List corresponding Required E_b/N_0 values for the coding scheme from Table B-2 in this row.

Code	Required E_b/N_0 (dB)	Max. Supported Symbol Rate (Msps)
LDPC (1024, 1/2)	2.03	300 Msps
LDPC (4096, 1/2)	1.33	300 Msps
LDPC (16384, 1/2)	1.01	300 Msps
LDPC (1024, 2/3)	2.91	300 Msps
LDPC (4096, 2/3)	2.13	300 Msps
LDPC (16384, 2/3)	1.81	300 Msps
LDPC (1024, 4/5)	3.96	300 Msps
LDPC (4096, 4/5)	3.13	300 Msps
LDPC (16384, 4/5)	2.79	300 Msps
LDPC (7136, 7/8)	3.96	300 Msps
Turbo (8920, 1/2)	1.12	3.2 Msps
Turbo (8920, 1/3)	0.48	4.8 Msps
Turbo (8920, 1/4)	0.23	6.4 Msps
Turbo (8920, 1/6)	-0.06	6.0 Msps

Table B-2: Required E_b/N_0 values for LDPC Coding Scheme at $BER = 10^{-7}$ assuming rectangular pulse shaping.

				DWE Forward Link at Max Distance (DSS-55 to MTN)		
	Node	Parameter	Value	Units	Equation	Comments
1	DSN	EIRP	109.5	dBW	X-band EIRP	DSN 820-100-H. DSN 34m BWG, 20 kW ⁶
2	Channel	Distance		km	Input, Orbit Dependent ⁷	
3	Channel	Center Frequency		MHz	Input, Antenna Dependent	
4	Channel	Free Space Loss		dB	$-[20*\log_{10}[(2)]+20*\log_{10}[(3)]+32.45]$	ITU-R P.525-5
5	Channel	Total Atmospheric Loss		dB	-0.5 dB at X-band	Atmospheric Loss at DSS-55, MDSCC ⁸
6	Channel	Other Losses		dB	Input, Antenna Dependent	Polarization/RFI Losses ⁹
7	MTN	Power Arriving at Antenna		dB	$(1)+(4)+(5)+(6)$	
8	MTN	Receive Losses		dB	Input, Antenna Dependent	e.g. Circuit Losses, Antenna Pointing Losses ⁹
9	MTN	S/C Antenna Efficiency		%	Input, Antenna Dependent	
10	MTN	S/C Antenna Diameter		m	Input, Antenna Dependent	
11	MTN	S/C Antenna Gain		dB	$10*\log_{10}[(9)*((10)*(3)*\pi*1000000/c)^2]$	
12	MTN	System Noise Temperature		dBK	Input, Antenna Dependent	
13	MTN	Receive G/T		dB/K	$(8)+(11)-(12)$	
14	MTN	Received P_r/N_0		dB-Hz	$(7)+(13)-10*\log_{10}[1.38E-23]$	K (Boltzmann's Constant) = 1.38e-23 W/K/Hz
15		Carrier Suppression		dB	Input, Losses for Carrier Synchronization	
16		Ranging Suppression		dB	Input, Losses for Simultaneous Ranging	
17	MTN	Received P_d/N_0		dB-Hz	$(14)-(15)-(16)$	
18	MTN	Implementation Loss		dB	Input, Receiver Dependent	
19		Receive Symbol Rate (Linear)		Msp/s	Input, Modulation Dependent ¹⁰	
20		Receive Symbol Rate (dB)		dB-Hz	$10*\log_{10}[(19)*1000000]$	
21		Coding Rate		-	Input Coding Rate ¹¹	
22		Receive Data Rate		Mbps	$(19)*(21)$	
23		Required E_b/N_0		dB	Input, Dependent on Coding Scheme ¹²	MTN-REQ-012: BER = 10 ⁻⁷
24		Required E_c/N_0		dB	$(23)+10*\log_{10}[(21)]$	
25		Required Margin	3	dB		MTN-REQ-011: Margin ≥ 3 dB
26		Excess Margin		dB	$(17)+(18)-(20)-(24)-(25)$	

Table B-3: DWE Forward Link

⁶Ka-band DWE Forward Link is not a threshold requirement but is a goal requirement for **MTN-REQ-041**. If needed for proposed configuration, use 103.8 dB for the DSN Ka-band EIRP. This value is taken from DSN 820-100-H for DSS-25, which has a 34m BWG at a transmit power of 300 W.

⁷Provide link budgets for both the minimum and maximum distances between Mars and Earth, as listed in **MTN-REQ-045**. Use 55,000,000 km for the minimum Mars-Earth Distance and 400,000,000 km for the maximum Mars-Earth Distance.

⁸Ka-band DWE Forward Link is not a threshold requirement but is a goal requirement for **MTN-REQ-041**. If needed for proposed configuration, use -5 dB for Ka-band Atmospheric Attenuation Values.

⁹List out all losses that covered within this box.

¹⁰In the uplink direction, DSN only modulates the symbols onto the carrier and radiates them. It is the mission operations center responsibility to deliver fully formed (including encoding) CCSDS CLTUs for radiation. The modulations compatible with DSN are documented in the DSN Link Design Handbook, 810-005 Module 205.

¹¹Select a DSN-compatible coding scheme and list the rate in this row.

¹²List corresponding Required E_b/N_0 values for the coding scheme from Table B-2 in this row.

Proximity Links

Advanced Proximity Return Link (Mars Surface to MTN)						
	Node	Parameter	Value	Units	Equation	Comments
1	Mars	EIRP	20	dBW		Nominal Case ¹³
2	Channel	Distance		km	Input, Orbit Dependent ¹⁴	
3	Channel	Center Frequency		MHz	Input, Antenna Dependent	
4	Channel	Free Space Loss		dB	$-[20*\log_{10}[(2)]+20*\log_{10}[(3)]+32.45]$	ITU-R P.525-5
5	Channel	Other Losses		dB	Input, Antenna Dependent	Polarization/RFI Losses ¹⁵
6	MTN	Power Arriving at Antenna		dB	$(1)+(4)+(5)$	
7	MTN	Receive Losses		dB	Input, Antenna Dependent	Circuit/Antenna Pointing Losses ¹⁵
8	MTN	S/C Antenna Efficiency		%	Input, Antenna Dependent	
9	MTN	S/C Antenna Diameter		m	Input, Antenna Dependent	
10	MTN	S/C Antenna Gain		dB	$10*\log_{10}[(8)*((9)*(3)*\pi*1000000/c)^2]$	
11	MTN	System Noise Temperature		dBK	Input, Antenna Dependent	
12	MTN	Receive G/T		dB/K	$(7)+(10)-(11)$	
13	MTN	Received P_r/N_0		dB-Hz	$(6)+(12)-10*\log_{10}[1.38E-23]$	K (Boltzmann's Constant) = $1.38e-23$ W/K/Hz
14	MTN	Implementation Loss		dB	Input, Receiver Dependent	
15		Receive Symbol Rate (Linear)		Msp/s	Input, Modulation Dependent	
16		Receive Symbol Rate (dB)		dB-Hz	$10*\log_{10}[(15)*1000000]$	
17		Coding Rate		-	Input Coding Rate ¹⁶	
18		Receive Data Rate		Mbps	$(15)*(17)$	
19		Required E_b/N_0		dB	Input, Dependent on Coding Scheme ¹⁷	MTN-REQ-012: BER = 10^{-7}
20		Required E_b/N_0		dB	$(19)+10*\log_{10}[(17)]$	
21		Required Margin	3	dB		MTN-REQ-011: Margin ≥ 3 dB
22		Excess Margin		dB	$(13)+(14)-(16)-(20)-(21)$	

Table B-4: Advanced Proximity Return Link

¹³The EIRP listed here is a nominal case used to compare link budget results across all vendors. This value is for illustrative purposes and should not be used as a constraint for design of the system.

¹⁴Provide a link budget with 600 km as a nominal case for comparison of link budget results across all vendors. While this value is in accordance with **MTN-REQ-027**, it is only an example for illustrative purposes and should not be used as a constraint for design of the system. Also provide link budgets for minimum and maximum distances to the surface based on proposed orbits.

¹⁵List out all losses that are covered within these rows.

¹⁶Select coding scheme and list the rate in this row.

¹⁷List corresponding Required E_b/N_0 values for the coding scheme in this row, either from Table B-2 as relevant or related value for proposed coding scheme.

Advanced Proximity Forward Link (MTN to Mars Surface)						
	Node	Parameter	Value	Units	Equation	Comments
1	MTN	Transmit Power		dBW	Input, Antenna Dependent	Circuit/Antenna Pointing Losses ¹⁸
2	MTN	Transmit Losses		dB	Input, Antenna Dependent	
3	MTN	S/C Antenna Efficiency		%	Input, Antenna Dependent	
4	MTN	S/C Antenna Diameter		m	Input, Antenna Dependent	
5	MTN	S/C Antenna Gain		dB	$10*\log_{10}[(3)*(4)*(8)*\pi*1000000/c]^2]$	
6	MTN	EIRP		dBW	(1)+(2)+(5)	
7	Channel	Distance		km	Input, Orbit Dependent ¹⁹	
8	Channel	Center Frequency		MHz	Input, Antenna Dependent	
9	Channel	Free Space Loss		dB	$-[20*\log_{10}(7)+20*\log_{10}(8)]+32.45]$	ITU-R P.525-5
10	Channel	Other Losses		dB	Input, Antenna Dependent	Polarization/RFI Losses ¹⁸
11	Mars	Power Arriving at Antenna		dB	(6)+(9)+(10)	
12	Mars	Receive G/T	-10	dB/K		Nominal Case ²⁰
13	Mars	Received P_r/N_0		dB-Hz	$(11)+(12)-10*\log_{10}[1.38E-23]$	K (Boltzmann's Constant) = 1.38e-23 W/K/Hz
14	Mars	Implementation Loss		dB	Input, Receiver Dependent	
15		Receive Symbol Rate (Linear)		Mbps	Input, Modulation Dependent	
16		Receive Symbol Rate (dB)		dB-Hz	$10*\log_{10}[(15)*1000000]$	
17		Coding Rate		-	Input Coding Rate ²¹	
18		Receive Data Rate		Mbps	$(15)*(17)$	
19		Required E_b/N_0		dB	Input, Dependent on Coding Scheme ²²	MTN-REQ-012: BER = 10^{-7}
20		Required E_r/N_0		dB	$(19)+10*\log_{10}[(17)]$	
21		Required Margin	3	dB		MTN-REQ-011: Margin ≥ 3 dB
22		Excess Margin		dB	$(13)+(14)-(16)-(20)-(21)$	

Table B-5: Advanced Proximity Forward Link

¹⁸List out all losses that are covered within these rows.

¹⁹Provide a link budget with 600 km as a nominal case for comparison of link budget results across all vendors. While this value is in accordance with **MTN-REQ-027**, it is only an example for illustrative purposes and should not be used as a constraint for design of the system. Also provide link budgets for minimum and maximum distances to the surface based on proposed orbits.

²⁰The G/T listed here is a nominal case used to compare link budgets across all vendors. This value is for illustrative purposes and should not be used as a constraint for design of the system.

²¹Select coding scheme and list the rate in this row.

²²List corresponding Required E_b/N_0 values for the coding scheme in this row, either from Table B-2 as relevant or related value for proposed coding scheme.

UHF Proximity Return Link (Mars Surface to MTN)						
	Node	Parameter	Value	Units	Equation	Comments
1	Mars	EIRP	3	dBW		Nominal Case ²³
2	Channel	Distance		km	Input, Orbit Dependent ²⁴	
3	Channel	Center Frequency		MHz	Input, Antenna Dependent	
4	Channel	Free Space Loss		dB	$-[20*\log_{10}(2)]+20*\log_{10}(3)+32.45]$	ITU-R P.525-5
5	Channel	Other Losses		dB	Input, Antenna Dependent	Polarization/RFI Losses ²⁵
6	MTN	Power Arriving at Antenna		dB	$(1)+(4)+(5)$	
7	MTN	Receive Losses		dB	Input, Antenna Dependent	Circuit/Antenna Pointing Losses ²⁵
8	MTN	S/C Antenna Gain		dB	Input, Antenna Dependent	
9	MTN	System Noise Temperature		dBK	Input, Antenna Dependent	
10	MTN	Receive G/T		dB/K	$(7)+(8)-(9)$	
11	MTN	Received P_r/N_0		dB-Hz	$(6)+(10)-10*\log_{10}[1.38E-23]$	K (Boltzmann's Constant) = 1.38e-23 W/K/Hz
12	MTN	Implementation Loss		dB	Input, Receiver Dependent	
13		Receive Symbol Rate (Linear)		Mbps	Input, Modulation Dependent	
14		Receive Symbol Rate (dB)		dB-Hz	$10*\log_{10}(13)*1000000]$	
15		Coding Rate		-	Input Coding Rate ²⁶	
16		Receive Data Rate		Mbps	$(13)*(15)$	
17		Required E_b/N_0		dB	Input, Dependent on Coding Scheme ²⁷	MTN-REQ-012: BER = 10^{-7}
18		Required E_s/N_0		dB	$(17)+10*\log_{10}[(15)]$	
19		Required Margin	3	dB		MTN-REQ-011: Margin ≥ 3 dB
20		Excess Margin		dB	$(11)+(12)-(14)-(18)-(19)$	

Table B-6: UHF Proximity Return Link

²³The EIRP listed here is a nominal case used to compare link budget results across all vendors. This value is for illustrative purposes and should not be used as a constraint for design of the system.

²⁴Provide a link budget with 600 km as a nominal case for comparison of link budget results across all vendors. While this value is in accordance with **MTN-REQ-027**, it is only an example for illustrative purposes and should not be used as a constraint for design of the system. Also provide link budgets for minimum and maximum distances to the surface based on proposed orbits.

²⁵List out all losses that are covered within these rows.

²⁶Select coding scheme and list the rate in this row.

²⁷List corresponding Required E_b/N_0 values for the coding scheme in this row, either from Table B-2 as relevant or related value for proposed coding scheme.

UHF Proximity Forward Link (MTN to Mars Surface)						
	Node	Parameter	Value	Units	Equation	Comments
1	MTN	Transmit Power		dBW	Input, Antenna Dependent	Circuit/Antenna Pointing Losses ²⁸
2	MTN	Transmit Losses		dB	Input, Antenna Dependent	
3	MTN	S/C Antenna Gain		dB	Input, Antenna Dependent	
4	MTN	EIRP		dBW	(1)+(2)+(3)	
5	Channel	Distance		km	Input, Orbit Dependent ²⁹	
6	Channel	Center Frequency		MHz	Input, Antenna Dependent	
7	Channel	Free Space Loss		dB	-[20*log10(5)]+20*log10(6)+32.45]	ITU-R P.525-5
8	Channel	Other Losses		dB	Input, Antenna Dependent	Polarization/RFI Losses ²⁸
9	Mars	Power Arriving at Antenna		dB	(4)+(7)+(8)	
10	Mars	Receive G/T	-25	dB/K		Nominal Case ³⁰
11	Mars	Received P _r /N ₀		dB-Hz	(9)+(10)-10*log10[1.38E-23]	K (Boltzmann's Constant) = 1.38e-23 W/K/Hz
12	Mars	Implementation Loss		dB	Input, Receiver Dependent	
13		Receive Symbol Rate (Linear)		Msp/s	Input, Modulation Dependent	
14		Receive Symbol Rate (dB)		dB-Hz	10*log10[(13)*1000000]	
15		Coding Rate		-	Input Coding Rate ³¹	
16		Receive Data Rate		Mbps	(13)*(15)	
17		Required E _b /N ₀		dB	Input, Dependent on Coding Scheme ³²	MTN-REQ-012: BER = 10 ⁻⁷
18		Required E _b /N ₀		dB	(17)+10*log10[(15)]	
19		Required Margin	3	dB		MTN-REQ-011: Margin ≥ 3 dB
20		Excess Margin		dB	(11)+(12)-(14)-(18)-(19)	

Table B-7: UHF Proximity Forward Link

²⁸List out all losses that are covered within these rows.

²⁹Provide a link budget with 600 km as a nominal case for comparison of link budget results across all vendors. While this value is in accordance with **MTN-REQ-027**, it is only an example for illustrative purposes and should not be used as a constraint for design of the system. Also provide link budgets for minimum and maximum distances to the surface based on proposed orbits.

³⁰The G/T listed here is a nominal case used to compare link budgets across all vendors. This value is for illustrative purposes and should not be used as a constraint for design of the system.

³¹Select coding scheme and list the rate in this row.

³²List corresponding Required E_b/N_0 values for the coding scheme in this row, either from Table B-2 as relevant or related value for proposed coding scheme.